Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Scatterplot of Relationship between Two Quantitative Variables WITH Simple Linear Regression Line

```
ggplot(<NAME OF DATASET>,
          aes(x = <NAME OF EV>,
              y = <NAME OF RV>))+
       geom_point()+
       geom_smooth(method = “lm”, se = F)
```

Creates a scatterplot with EV on the x-axis and RV on the y-axis along with the simple linear regression line

Scatterplot of Relationship between Two Quantitative Variables WITH Color & Shape by Third Categorical Variable

```
ggplot(<NAME OF DATASET>,
          aes(x = <NAME OF QUANTITATIVE EV>,
              y = <NAME OF RV>))+
       geom_point(aes(shape = <NAME OF CATEGORICAL EV>,
                      color = <NAME OF CATEGORICAL EV>))
```

Creates a scatterplot with a quantitative EV on the x-axis and RV on the y-axis, with the colors and shapes of points corresponding to a categorical EV.

Scatterplot of Relationship between Two Quantitative Variables and 1 Cat Variable WITH Parallel Regression Lines (no interaction)

```
library(moderndive)
ggplot(<NAME OF DATASET>,
          aes(x = <NAME OF QUANTITATIVE EV>,
              y = <NAME OF RV>))+
       geom_point(aes(shape = <NAME OF CATEGORICAL EV>,
                      color = <NAME OF CATEGORICAL EV>))+
       geom_parallel_slopes(aes(color = <NAME OF CATEGORICAL EV>), se = F)
```

Creates a scatterplot with a quantitative EV on the x-axis and RV on the y-axis, with the colors and shapes of points corresponding to a categorical EV AND parallel regression lines for groups (for model with no interaction). You need the moderndive package to be loaded for this function.

Scatterplot of Relationship between Two Quantitative Variables and 1 Cat Variable WITH Different Slope Regression Lines (Interaction)

```
ggplot(<NAME OF DATASET>,
          aes(x = <NAME OF QUANTITATIVE EV>,
              y = <NAME OF RV>))+
       geom_point(aes(shape = <NAME OF CATEGORICAL EV>,
                      color = <NAME OF CATEGORICAL EV>))+
       geom_smooth(aes(color = <NAME OF CATEGORICAL EV>),
                       method = “lm”, se = F)
```

Creates a scatterplot with a quantitative EV on the x-axis and RV on the y-axis, with the colors and shapes of points corresponding to a categorical EV AND parallel regression lines for groups (for model with no interaction).

Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Fit a Regression Model and See Model Output

```
mymod <- lm(<NAME OF RV> ~ <NAME OF EV1> + <NAME OF EV2> + …,
                data = <NAME OF DATASET>)
summary(mymod)
```

Formula note - use the following syntax to add an interaction: `<NAME OF EV1>:<NAME OF EV2>`

Output will be the estimated coefficients for the model and other information like R^2 etc.

Confidence Intervals for Regression Coefficients

Assuming you have fit a model like above called **`mymod`**

```
confint(mymod, level = 0.95)
```

Output will be the lower and upper bounds for a 95% confidence interval for every coefficient. You can change the level to get different X% confidence intervals.

Partial F-Test

Assuming you have fit a regression model **bigmod** (like above) and a model **smallmod** that removed terms from the "bigger" model:

```
anova(bigmod, smallmod)
```

Output will be the F-statistic and p-value for the test of whether at least one of the additional coefficients in the bigger model not zero.

Scatterplot of Residuals versus Predicted Values

Assuming you have fit a model like above using lm() called **`mymod`**

```
moddat <- mymod |>
  augment()

ggplot(moddat) +
  geom_hline(yintercept = 0, color = "blue") +
  geom_point(aes(x = .fitted, y = .resid)) +
  labs(x = "Predicted Values",
       y = "Residual")
```

Creates a plot of residuals versus predicted values from a model with a blue line at 0.

Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Histogram of Residuals

Assuming you have fit a model like above using lm() called **mymod,** and saved a dataset using augment() like **moddat** above:

```
ggplot(moddat, aes(x = .resid)) +
  geom_histogram() +
  labs(x = "Residual",
       y = "Count",
       title = "Distribution of Residuals")
```

Creates a histogram of residuals from a model.